Comprehensive Structural and Electrical Analysis of Scaled Nanosheet FETs for Advanced Technology Nodes using Sentaurus TCAD

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***Abstract*—This study presents the physics-based design and numerical simulation of a gate-all-around (GAA) nanosheet field-effect transistor (NS-FET) using the Synopsys Sentaurus TCAD platform, targeting sub-10-nm technology node applications. A three-dimensional device structure comprising vertically stacked silicon nanosheets was constructed, and a calibrated mesh strategy was employed to resolve critical electrostatic and quantum-mechanical effects. The simulation framework incorporates advanced physical models, including drift–diffusion transport, density-gradient quantum corrections, high-field mobility degradation, carrier scattering mechanisms, and interface trap dynamics. Key electrical metrics—such as the transfer characteristics, output characteristics, threshold voltage, subthreshold swing, drain-induced barrier lowering (DIBL), and on-state current—were extracted to evaluate device performance. The results indicate that the nanosheet architecture achieves enhanced gate controllability and significant suppression of short-channel effects relative to conventional FinFET designs.**

***Keywords*—Nanosheet FET, Gate-All-Around (GAA) Transistor, Quantum Confinement, Carrier Mobility Modeling**

I. Introduction

Continuous scaling of metal–oxide–semiconductor field-effect transistors (MOSFETs) has been the driving force behind advancements in integrated circuit technology. However, as device dimensions approach the 7nm technology node, conventional planar and FinFET architectures encounter fundamental limitations, including degraded electrostatic control, increased short-channel effects (SCEs), variability, and insufficient drive current. To overcome these challenges, gate-all-around (GAA) transistor structures have emerged as leading candidates for next-generation CMOS technology. Among various GAA configurations, the nanosheet field-effect transistor (NS-FET) has attracted significant attention for its enhanced gate controllability and flexibility in tuning the effective channel width through sheet stacking. Nanosheet structures maintain a larger cross-sectional area, enabling higher drive current while still providing superior electrostatic integrity.

Technology computer-aided design (TCAD) tools, such as Synopsys Sentaurus, play a critical role in evaluating and optimizing nanosheet device performance before fabrication. TCAD enables rigorous, physics-based simulation that incorporates quantum effects, mobility degradation, carrier transport models, and interface-trap dynamics. Through accurate numerical modelling, researchers can investigate design trade-offs, optimize channel geometry, assess short-channel behaviour, and quantify electrical characteristics under various bias conditions.

II. Device Design and Simulation Methodology

The nanosheet field-effect transistor (NS-FET) was designed as a three-dimensional gate-all-around (GAA) architecture to ensure superior electrostatic control and reduced short-channel effects. The device consists of vertically stacked silicon nanosheets separated by thin oxide layers to achieve an enhanced effective channel width while maintaining strong gate coupling. Each nanosheet serves as an independent conductive channel fully surrounded by the gate dielectric, metal gate and TiN capping layers. The source and drain regions were heavily doped to facilitate low-resistance carrier injection, while the channel region remained lightly doped or undoped to minimize threshold voltage variability and improve carrier mobility. Key structural parameters such as nanosheet thickness(T_{ch}), width(W_{ch}), inter-sheet spacing(T_{sp}), gate length(Lg), and spacer dimensions(L_{sp}) were defined based on typical 7nm technology node requirements. A high-K dielectric material was employed as the gate insulator to reduce gate leakage.

TABLE I: Design parameters for this 7nm Nanosheet FET [3] [6]

Parameters	Symbol	Value
Length of the device	$2L_{sd} + 2L_{sp} + L_g$	50 nm
Height of the device	$Hn + h$	120 nm
Width of the device	Wn	40 nm
PTS region height	h_{PTS}	30 nm
Length of the Spacer	L_{sp}	7 nm
Length of the S/D region	L_{sd}	10 nm
Width of the Channel	W_{ch}	26 nm
Oxide thickness	t_{ox}	2 nm
Channel doping	Ch_Dop	$1.0 \times 10^{15} cm^{-3}$
Source / Drain doping	SD_Dop	$1.0 \times 10^{20} cm^{-3}$

Doping profiles were specified for the source, drain, and channel regions to achieve the desired device characteristics. A high-k gate dielectric (HfO_2) and metal gate (polysilicon) were employed to enhance gate control and reduce leakage currents. To analyse gate leakage, SiO_2 is also used as the gate dielectric. The 3D device mesh was carefully generated

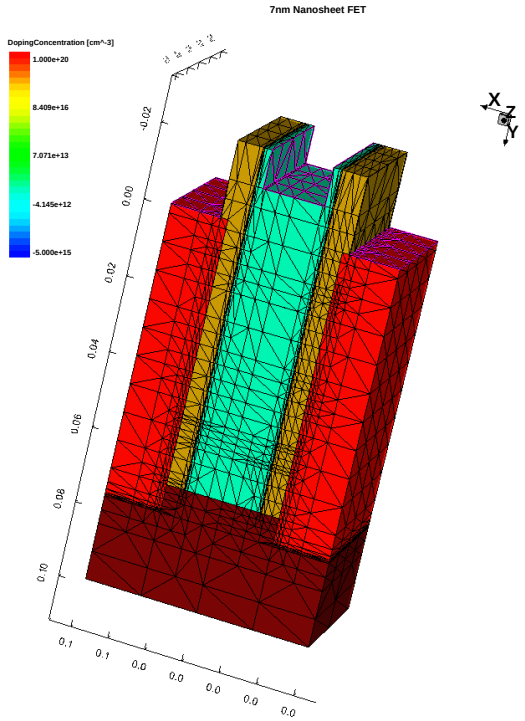


Fig. 1: 7nm Nanosheet FET structure with meshing

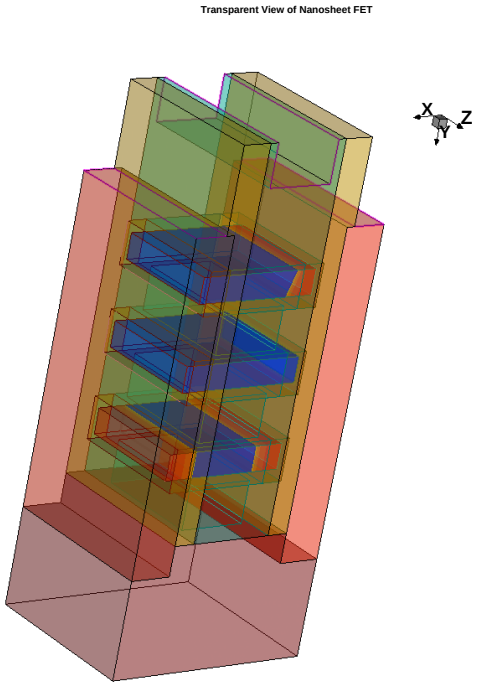


Fig. 3: Transparent 3D view of 7nm Nanosheet FET

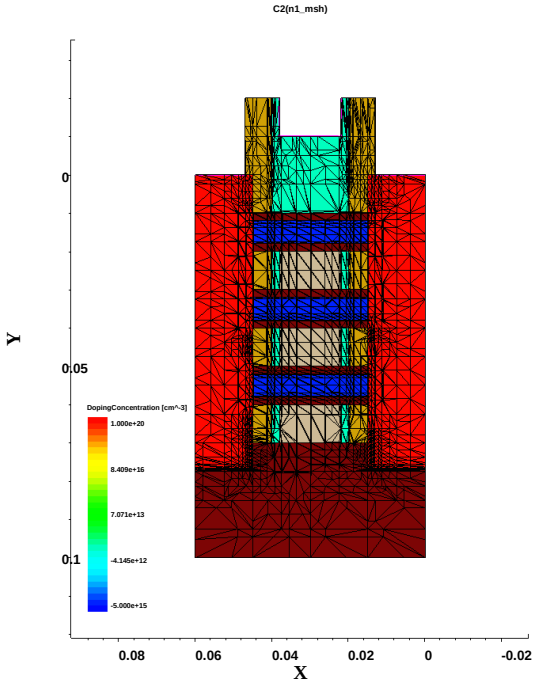


Fig. 2: Cross-section (Z-plane) view of 7nm Nanosheet FET with meshing

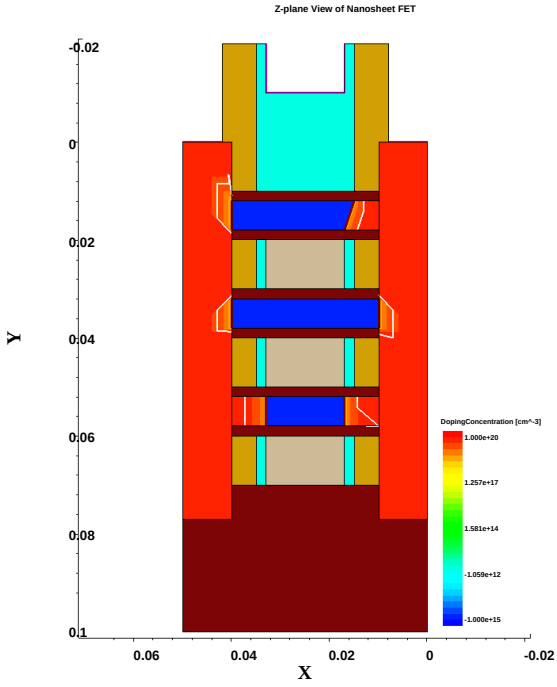
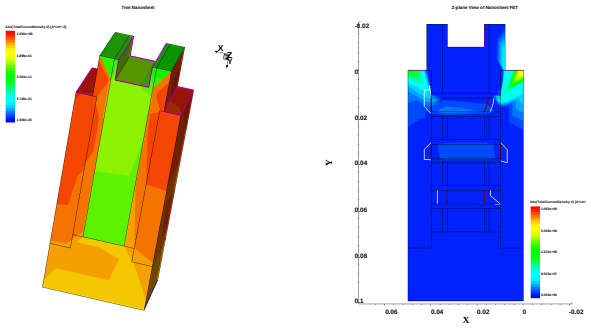


Fig. 4: Doping Concentration(2D, z-plane) at every section of 7nm Nanosheet FET

to ensure numerical accuracy during simulation. (* All the codes are available in the appendix section)

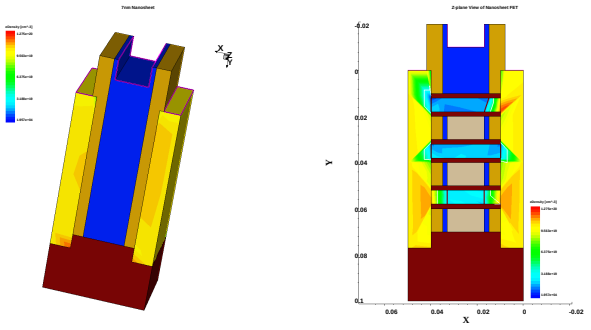
The various types of property variation (at $V_{gs} = 1.2V$ and $V_{ds} = 1V$) -

Electrical simulations were carried out using Sentaurus Device (SDevice) under various bias conditions to obtain the transfer ($I_d - V_g$) and output ($I_d - V_d$) characteristics. Physical



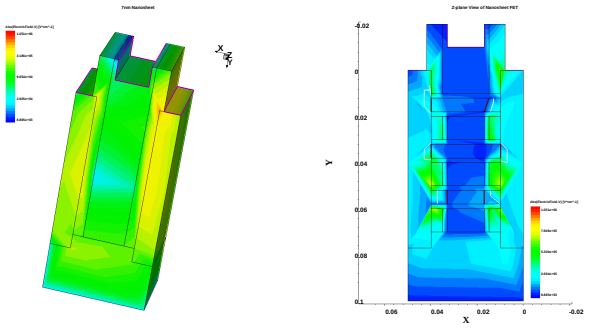
(a) $3D$ view (b) $2D$ view(z-plane)

Fig. 5: Total current Density variation on $7nm$ Nanosheet FET



(a) $3D$ view (b) $2D$ view(z-plane)

Fig. 6: Electron Density variation on $7nm$ Nanosheet FET



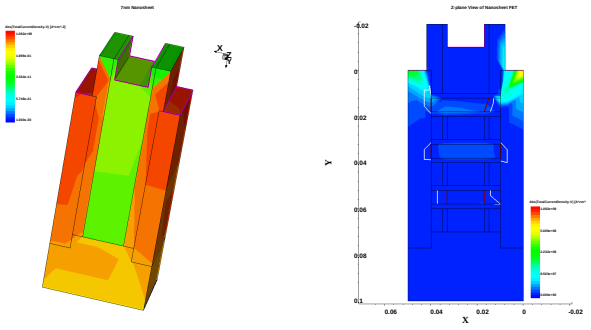
(a) $3D$ view (b) $2D$ view(z-plane)

Fig. 7: Electric field variation on $7nm$ Nanosheet FET

models such as mobility degradation, quantum confinement, and recombination effects were included to capture realistic device behaviour.

III. Simulation Results and Analysis

The simulated Nanosheet FET device demonstrated excellent electrostatic control and superior performance compared to conventional planar MOSFETs. The transfer characteristics $(I_d - V_g)$ showed a sharp turn-on behaviour, indicating strong gate control and a low subthreshold swing.



(a) $3D$ view (b) $2D$ view(z-plane)

Fig. 8: Electrostatic Potential variation on $7nm$ Nanosheet FET

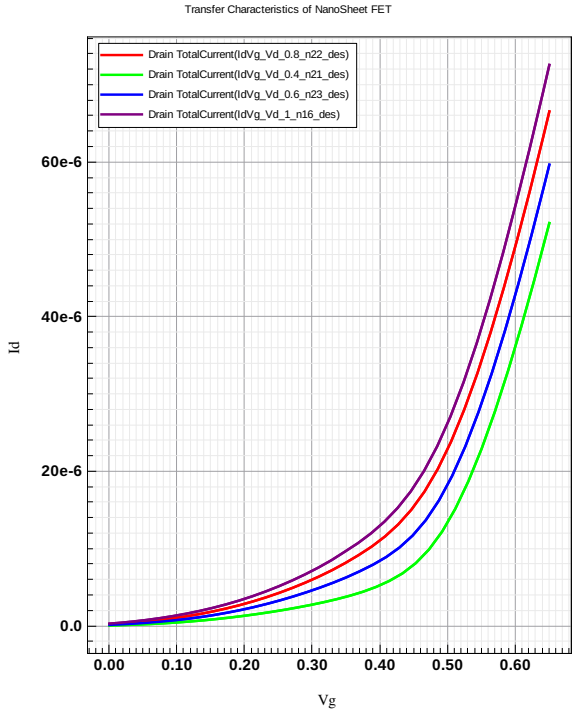


Fig. 9: Transfer Characteristics $(I_d \ vs \ V_g)$ curve of Nanosheet FET (with HfO_2 gate Oxide) at different V_{ds}

Physics Model

Hydrodynamic Model [5]: In the hydrodynamic model, the current density for electrons is given by -

$$\overrightarrow{J_n} = \mu_n\Big(n.\nabla E_C + kT_n\nabla n - nkT_n\nabla ln\gamma_n + \lambda_n f_n^{td} kn\nabla T_n - 1.5nkT_n.\nabla[lnm_n]\Big) \quad (1)$$

The current density for holes is given by -

$$\overrightarrow{J_p} = \mu_p\Big(p.\nabla E_V + kT_p\nabla p - pkT_p\nabla ln\gamma_p + \lambda_p f_p^{td} kp\nabla T_p - 1.5pkT_p.\nabla[lnm_p]\Big) \quad (2)$$

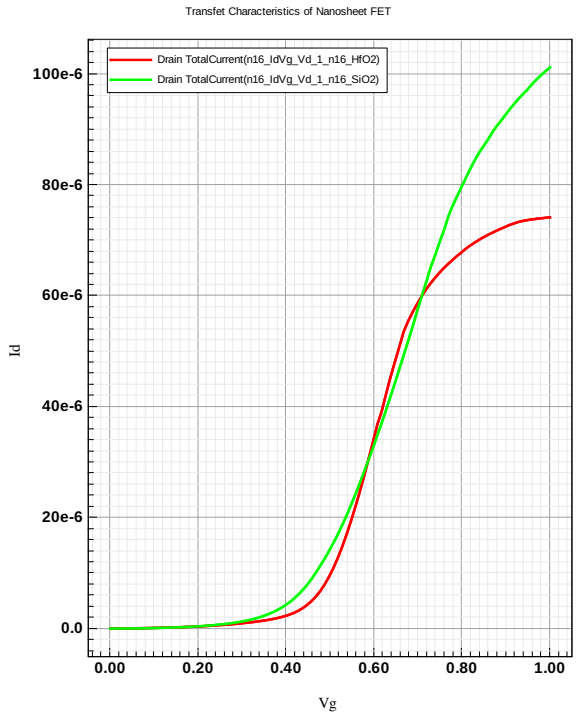


Fig. 10: Transfer Characteristics (I_d vs V_g) curve of Nanosheet FET (with SiO_2 and HfO_2 dielectric) at $V_{ds} = 1V$

The first term takes into account the contribution due to the spatial variations of electrostatic potential, electron affinity, and the band gap. The second, third, fourth and fifth terms take into account the contribution due to the gradient of concentration, the carrier temperature gradients, and the spatial variation of the effective masses m_p & m_n respectively. For Fermi-Boltzmann statistics, $\gamma_n = \gamma_p = \lambda_n = \lambda_p = 1$. The thermal diffusion constants are f_n^{td} and f_p^{td}.

Density-gradient quantization model [5]: For the density-gradient model, the modified potential will be

$$\bar{\Phi} = E_C + \phi_m + \Lambda_n \tag{3}$$

where E_C = Conduction band energy in eV, $\phi_m = -k_B T.ln(\frac{N_C}{n_i})$
The Λ_n is given by this partial differentiation equation.

$$\Lambda_n = -\frac{\gamma\hbar^2}{12m_n}\left[\nabla^2 ln(n) + \frac{1}{2}(\nabla lnn)^2\right] \tag{4}$$

where γ is a fitting parameter.

- A quantum transport model that includes quantum mechanical effects in the device simulation.
- It uses the density gradient quantum potential to model the wave-like behaviour of carriers in confined regions, like transistor channels.
- Provides a more physically accurate solution for carrier quantization compared to classical transport models.

Direct Tunneling Model: [5] In Sentaurus, the direct tunneling model presents the formulas for electrons only; the hole expressions are analogous. The electron tunneling current density is:

$$\vec{J_n} = \frac{qm_C k}{2\pi^2\hbar^3}\int_0^\infty dE.Y(E)\left[T(0).ln\left(e^{\frac{E_F(0)-E_C(0)-E}{kT(0)}}+1\right) - T(d)ln\left(e^{\frac{E_F(d)-E_C(0)-E}{kT(d)}}+1\right)\right] \tag{5}$$

where d is the effective thickness of the barrier, m_C is a mass prefactor, the argument "0" denotes one (the 'substrate') side of the barrier and "d" denotes the other ('gate') side and E is the energy of the elastic tunnel process[related to $E_C(0)$]. $Y(E)$ is the transmission coefficient for a trapezoidal potential barrier.

Carrier–Carrier Scattering: [5] The carrier–carrier contribution to the overall mobility degradation is captured in the mobility term μ_{eh}. This is combined with the mobility contributions from other degradation models(μ_{other}), according to Matthiessen's rule:

$$\frac{1}{\mu} = \frac{1}{\mu_{other}} + \frac{1}{\mu_{eh}} \tag{6}$$

The model also provides a formula for the scattering rate (inverse of relaxation time) as a function of the carrier energy and the doping concentration.

Schokley-Read-Hall recombination model [2]: Recombination through deep defect levels in the gap is normally referred to as Schokley-Read-Hall (SRH) recombination. In the Sentaurus device, the following term is implemented -

$$R_{net}^{SRH} = \frac{np - n_{i,eff}^2}{\tau_p\,(n+n_1) + \tau_n\,(p+p_1)} \tag{7}$$

where,

$$n_1 = n_{i,eff}.\exp\left(\frac{E_{trap}}{KT}\right) \tag{8}$$

and

$$p_1 = n_{i,eff}.\exp\left(\frac{-E_{trap}}{KT}\right) \tag{9}$$

E_{trap} is the difference between the defect level and the intrinsic level.

Modified Local-Density Approximation [4]: In brief, the model replaces the bulk density of states (DOS) in a semiconductor with a position-dependent local DOS field. The difference between the bulk DOS and the modified local DOS reflects the effect of applying specific interface boundary conditions to the carrier wave functions. While the six-band $K.p$ Schrodinger equation provides a general model for hole quantum effects in silicon, the large computational effort and the nonlocal character of this approach are not well suited to routine TCAD device simulation. In contrast, quantum correction models based on the MLDA model provide good

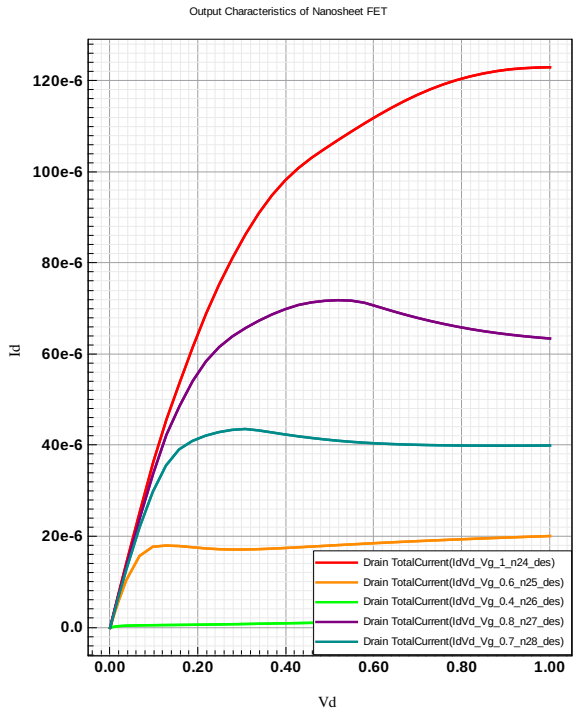


Fig. 11: Output Characteristics ($I_d \ \ vs \ \ V_d$) of Nanosheet FET (with HfO_2 gate-dielectric) at Different V_{gs}

computational efficiency and numerical convergence. For ellipsoidal bands, derives an analytical expression for the DOS of carriers near the interface -

$$D(\varepsilon, z) = \frac{1}{\pi^2\hbar^2}\sqrt{\frac{2m_1m_2m_3}{\hbar^2}\varepsilon}\left[1 - j_0\left(2z\sqrt{\frac{2}{w_{33}\hbar^2}\varepsilon}\right)\right] \tag{7}$$

where ϵ is the carrier energy, z is a distance from the interface, m_i are the effective masses along principal axes $\bar{e_l}$ of the ellipsoid, and j_0 is the spherical Bessel function $j_0 = (\lambda)(1/\lambda)sin\lambda$. The reciprocal mass component w_{33} is given by the expression $w_{33} = \sum_j (1/m_j)(e_3\bar{e_j})^2$, where e_3 is the interface normal vector (together with two in-plane unit vectors e_1 and e_2, this defines the interface coordinate system).

Band-to-Band Model [5]: Band-to-band tunnelling is controlled by the Band2Band option of Recombination. The modified electron density reads -

$$\bar{n} = n\left(\frac{n_{i,eff}}{N_C}\right)^{\frac{\gamma_n|\nabla E_{F,n}|}{F}} \tag{8}$$

there is a similar relation for $\bar{p}$ And $\gamma_n = \frac{n}{n+n_{eff}}$. This is called Schenk Density Correction, where n_{eff} is the reference density of e^-.

Fermi Model [5]: Boltzmann statistics is assumed for electrons and holes. Physically more correct, Fermi (also called Fermi–Dirac) statistics is used here. Fermi statistics becomes important for high values of carrier densities where -

$$n = \gamma_n N_C . e^{\frac{E_{F,n}-E_C}{K_B T}} \tag{9}$$

$$p = \gamma_p N_V . e^{\frac{E_V-E_{F,n}}{K_B T}} \tag{10}$$

Auger Recombination [5]: The rate of band-to-band Auger recombination is given by -

$$R^A_{net} = (nC_n + pC_p)(np - n^2_{i,eff}) \tag{11}$$

Auger recombination is typically important at high carrier densities.

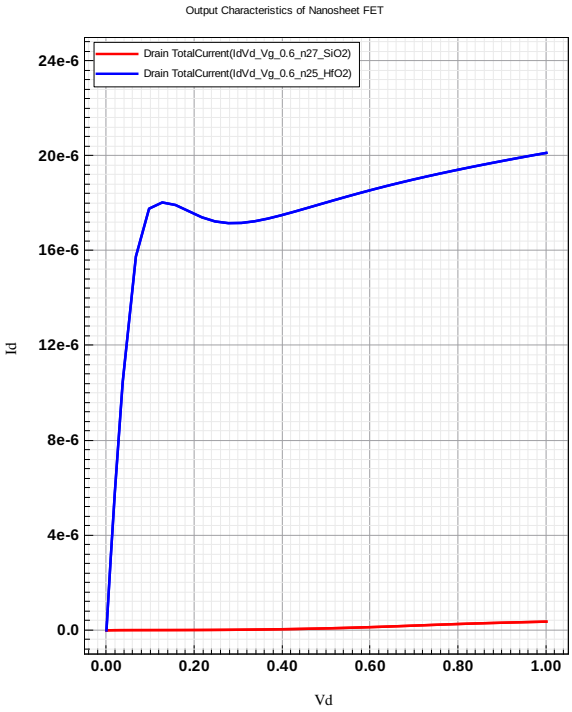


Fig. 12: $I_d \ \ vs \ \ V_d$ graph of Nanosheet for various gate dielectric (SiO_2 and HfO_2) at $V_{gs} = 0.6V$

TABLE II: The various gate metal work function and their corresponding off-current

Gate Material	Workfunction	Off-current	V_{th}
PolySilicon	4.1 eV	$3.433 \times 10^{-07}A$	$0.528V$
Aluminum(Al)	4.28 eV	$8.821 \times 10^{-06}A$	$0.068V$
Titanium(Ti)	4.33 eV	$9.313 \times 10^{-08}A$	$0.218V$
Molybdenum(Mo)	4.6 eV	–	$0.507V$

The output characteristics ($I_d - V_d$) confirmed good current saturation and high drive current capability.

Key performance parameters were extracted from the simulation results. The threshold voltage is $0.4558V$ (for PolySilicon and HfO_2 Gate Oxide).

A. Key Observation

- From figure 10 at $V_{ds} = 1V$ and $V_{gs} = 1V$, $I_{d,SiO_2} > I_{d,HfO_2}$, it is due to in case of SiO_2 gate oxide, there is a significant amount of leakage current, which increases the overall drain current.
- In figure 11, the I_d current first increases and then slightly decreases and saturates, this is because of scattering

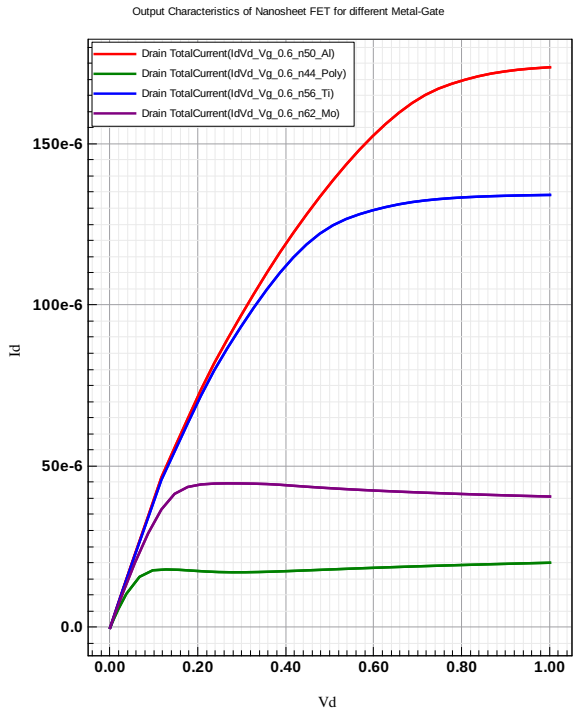


Fig. 13: Output Characteristics (I_d vs V_d) graphs of Nanosheet FET for various gate material workfunction(PolySi, Al, Ti and Mo)

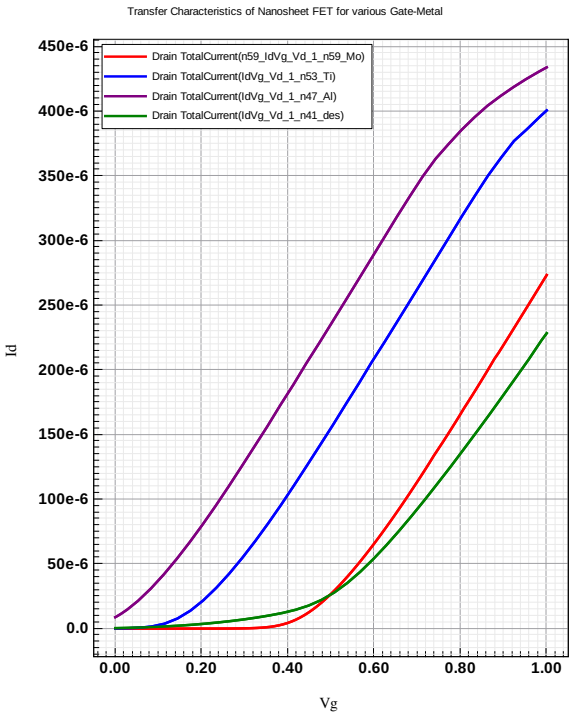


Fig. 14: Transfer Characteristics (I_d vs V_g) graphs of Nanosheet FET for various gate material workfunction(PolySi and Mo)

effects, i.e Coulomb, Phonon and surface roughness scattering.

- In figure 10, I_d current due to HfO_2 gate oxide saturates early and makes a curve diagram after the linear region. Due to surface roughness between HfO_2 and Si, the effective mobility (μ_{eff}) of e^- got decreased, and the current saturates.
- In figure 13, Polysilicon gives more resistance, that's why, as a metal gate, it shows the lowest I_d than Al, Ti and Mo.

IV. Conclusion

Through detailed simulations of the Nanosheet by incorporating advanced physical models—such as mobility degradation, quantum confinement, and interface trap dynamics—the electrical behaviour of the device was thoroughly evaluated. The obtained transfer and output characteristics reveal that the nanosheet FET offers high drive current, low subthreshold swing, and excellent suppression of short-channel effects, including minimal drain-induced barrier lowering. The fully surrounded gate geometry ensures strong channel confinement, reduced leakage, and stable saturation behaviour across a wide range of operating conditions. These performance advantages confirm the nanosheet FET as a robust candidate for technology scaling beyond the 7nm node. Overall, the study validates the nanosheet transistor's potential for integration into future high-performance and low-power CMOS technologies. The results presented here provide valuable insight into device behaviour, enabling further optimization of structural parameters, material choices, and process conditions for next-generation semiconductor applications.

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APPENDIX A
SDE: 7NM NANOSHEET FET DEVICE STRUCTURE CODE

```
(sde:clear)

;;;;;;;;;;;;;;; Dimensions ;;;;;;;;;;;;

;;;;; Total length of Nanosheet, L = 2Lsd+2Lsp + Lg ;;;;;;
;;;;; Gate length ;;;;;;;;
(define Lg (* 0.001 @Lg@))
;;;;; Total width ;;;;
(define Wn (* 0.001 @Wn@))
;;;;; Total Height ;;;;;;
(define Hn (* 0.001 @Hn@))
;;;;;; Extra Spacer Height from Base position ;;;;;;;
(define h (* 0.001 @h@))
;;;;;; PTS region height from lowest point ;;;;;;;;;
(define h_PTS (* 0.001 @h_PTS@))
;;;;;;; S/D depth in PTS region ;;;;;;
(define d_PTS (* 0.001 @d_PTS@))
;;;;;;; Length of Spacer ;;;;;;
(define Lsp (* 0.001 @Lsp@))
;;;;;;; Length of S/D region ;;;;;;
(define Lsd (* 0.001 @Lsd@))
;;;;;;;; Gate Oxide thicknes ;;;;;;
(define tox (* 0.001 @tox@))
;;;;;;;; Channel width ;;;;;;;;;;;
(define Wch (* 0.001 @Wch@))
;;;;;;;; TiN Block width ;;;;;;;;;
(define Tsp (* 0.001 @Tsp@))
;;;;;;;; Gate region thickness ;;;;;;
(define Tg (* 0.001 @Tg@))


;;;;;;;;;;;;;;;; Doping Variable ;;;;;;;;;;;
(define Ch_Dop @Ch_Dop@)
(define SD_Dop @SD_Dop@)

;;;;;;;;;;;;;;;;;;;;;;;;;;;Oxide Base;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
(sdegeo:create-cuboid (position 0 0 0) (position (+ (+ (* Lsd 2) (* Lsp 2)) Lg) Hn
    Wn) "SiO2" "Oxide")

;;;;;;;;;;;;; Spacer Block ;;;;;;;;;;;;;
(sdegeo:create-cuboid (position (- Lsd 0.002) (- 0 h) 0) (position (+ (+ (+ Lsd (*
    Lsp 2)) Lg) 0.002) (- Hn h_PTS) Wn) "Si3N4" "Spacer")

;;;;;;;;;;;; Source Region ;;;;;;;;;;;;
(sdegeo:create-cuboid (position 0 0 0) (position Lsd (+ (- Hn h_PTS) d_PTS) Wn)
    "Silicon" "Source_region")

;;;;;;;;;;;; Drain Region ;;;;;;;;;;;;
(sdegeo:create-cuboid (position (+ (+ Lsd (* Lsp 2)) Lg) 0 0) (position (+ (+ (* Lsd
    2) (* Lsp 2)) Lg) (+ (- Hn h_PTS) d_PTS) Wn) "Silicon" "Drain_region")

;;;;;;;;;;;; Gate material ;;;;;;;;;;;
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```
(sdegeo:create-cuboid (position (- (+ Lsd Lsp) 0.002) (- 0 h) 0) (position (+ (+ (+ Lsd Lsp) Lg) 0.002) (- Hn h_PTS) Wn) "PolySilicon" "Gate_material")

;;;;;;;;;;; Air Section ;;;;;;;;;;;;
(sdegeo:create-cuboid (position (+ Lsd Lsp) (- 0 h) 0) (position (+ (+ Lsd Lsp) Lg) (- 0 (* h 0.5)) Wn) "Vacuum" "Vacuum_section")

;;;;;;;;;;; TiN box design ;;;;;;;;
(sdegeo:create-cuboid (position (+ Lsd Lsp) (- Hn (+ h_PTS (+ (* Tsp 3) (* Tg 2)))) (* (- Wn Wch) 0.5)) (position (+ (+ Lsd Lsp) Lg) (- Hn h_PTS) (* (+ Wn Wch) 0.5)) "TiN" "Capping_layer")

;;;;;;;;;;; 1st Channel Oxide Formation ;;;;;;;;;
(sdegeo:create-cuboid (position Lsd (- Hn (+ h_PTS (+ Tsp Tg))) (* (- Wn (+ Wch (* tox 2))) 0.5)) (position (+ Lsd (+ Lg (* Lsp 2))) (- Hn (+ h_PTS Tsp)) (* (+ (+ Wn Wch) (* tox 2)) 0.5)) "HfO2" "Gate_Oxide")

;;;;;;;;;;; Channel 1 ;;;;;;;;;;;;;;
(sdegeo:create-cuboid (position Lsd (+ (- (- (- Hn h_PTS) Tsp) Tg) tox) (* (- Wn Wch) 0.5)) (position (+ Lsd (+ Lg (* Lsp 2))) (- (- (- Hn h_PTS) Tsp) tox) (* (+ Wn Wch) 0.5)) "Silicon" "first_Channel")

;;;;;;;;;; 2nd Channel Oxide Formation ;;;;;;;;
(sdegeo:create-cuboid (position Lsd (- Hn (+ h_PTS (+ (* Tsp 2) (* Tg 2)))) (* (- Wn (+ Wch (* tox 2))) 0.5)) (position (+ Lsd (+ Lg (* Lsp 2))) (- Hn (+ h_PTS (+ (* Tsp 2) Tg))) (* (+ (+ Wn Wch) (* tox 2)) 0.5)) "HfO2" "Gate_Oxide")

;;;;;;;;;; Channel 2 ;;;;;;;;;;;;
(sdegeo:create-cuboid (position Lsd (+ (- (- (- Hn h_PTS) (* Tsp 2)) (* Tg 2)) tox) (* (- Wn Wch) 0.5)) (position (+ Lsd (+ Lg (* Lsp 2))) (- (- (- (- Hn h_PTS) (* Tsp 2)) Tg) tox) (* (+ Wn Wch) 0.5)) "Silicon" "second_Channel")

;;;;;;;;; 3rd Channel Oxide Formation ;;;;;;;;;
(sdegeo:create-cuboid (position Lsd (- Hn (+ h_PTS (+ (* Tsp 3) (* Tg 3)))) (* (- Wn (+ Wch (* tox 2))) 0.5)) (position (+ Lsd (+ Lg (* Lsp 2))) (- Hn (+ h_PTS (+ (* Tsp 3) (* Tg 2)))) (* (+ (+ Wn Wch) (* tox 2)) 0.5)) "HfO2" "Gate_Oxide")

;;;;;;;;; Channel 3 ;;;;;;;;;;;;;
(sdegeo:create-cuboid (position Lsd (+ (- (- (- Hn h_PTS) (* Tsp 3)) (* Tg 3)) tox) (* (- Wn Wch) 0.5)) (position (+ Lsd (+ Lg (* Lsp 2))) (- (- (- (- Hn h_PTS) (* Tsp 3)) (* Tg 2)) tox) (* (+ Wn Wch) 0.5)) "Silicon" "third_Channel")



;;;;;;;;;;;;;;;;; Doping Layers ;;;;;;;;;;;;;
(sdedr:define-constant-profile "Const.Source_region" "ArsenicActiveConcentration" @SD_Dop@ )
(sdedr:define-constant-profile-region "PlaceCD.Source_region" "Const.Source_region" "Source_region" )

(sdedr:define-constant-profile "Const.Drain_region" "ArsenicActiveConcentration" @SD_Dop@ )
(sdedr:define-constant-profile-region "PlaceCD.Drain_region" "Const.Drain_region" "Drain_region" )

```

```
84  (sdedr:define-constant-profile "Const.first_Channel" "BoronActiveConcentration"
        @Ch_Dop@ )
85  (sdedr:define-constant-profile-region "PlaceCD.first_Channel" "Const.first_Channel"
        "first_Channel" )
86  (sdedr:define-constant-profile "Const.second_Channel" "BoronActiveConcentration"
        @Ch_Dop@ )
87  (sdedr:define-constant-profile-region "PlaceCD.second_Channel"
        "Const.second_Channel" "second_Channel" )
88  (sdedr:define-constant-profile "Const.third_Channel" "BoronActiveConcentration"
        @Ch_Dop@ )
89  (sdedr:define-constant-profile-region "PlaceCD.third_Channel" "Const.third_Channel"
        "third_Channel" )
90
91  ;;;;;;;;;;;; Source Contact ;;;;;;;;;;;
92  (sdegeo:define-contact-set "Source" 6 (color:rgb 1 0 0 ) "##")
93  (sdegeo:set-current-contact-set "Source")
94  (sdegeo:define-3d-contact (list (car (find-face-id (position (* Lsd 0.5) 0 (* Wn
        0.5))))) "Source")
95
96  ;;;;;;;;;;;; Drain Contact ;;;;;;;;;;;;
97  (sdegeo:define-contact-set "Drain" 6 (color:rgb 0 0 1 ) "##")
98  (sdegeo:set-current-contact-set "Drain")
99  (sdegeo:define-3d-contact (list (car (find-face-id (position (+ (+ (* Lsp 2) Lg) (*
        Lsd 1.5)) 0 (* Wn 0.5))))) "Drain")
100
101 ;;;;;;;;;;;;;; Gate Contact ;;;;;;;;;;;;;
102 (sdegeo:define-contact-set "Gate" 6 (color:rgb 0 1 0 ) "##")
103 (sdegeo:set-current-contact-set "Gate")
104 (sdegeo:define-3d-contact (list (car (find-face-id (position (+ Lsd (+ Lsp (* Lg
        0.5))) (- 0 (* h 0.5)) (* Wn 0.5))))) "Gate")
105 (sdegeo:define-3d-contact (list (car (find-face-id (position (+ Lsd Lsp) (* (- 0 h)
        0.75) (* Wn 0.5))))) "Gate")
106 (sdegeo:define-3d-contact (list (car (find-face-id (position (+ Lsd (+ Lsp Lg)) (*
        (- 0 h) 0.75) (* Wn 0.5))))) "Gate")
107
108 ;;;;;;;;;;;; Meshing ;;;;;;;;;;;;;
109 (sdedr:define-refeval-window "RefEvalWin_1" "Cuboid" (position 0 (- 0 h) 0)
        (position (+ (+ (* Lsd 2) (* Lsp 2)) Lg) Hn Wn))
110 (sdedr:define-refinement-size "RefinementDefinition_1" 0.05 0.05 0.05 0.01 0.01 0.01
        )
111 (sdedr:define-refinement-placement "RefinementPlacement_1" "RefinementDefinition_1"
        (list "window" "RefEvalWin_1" ) )
112 (sdedr:define-refinement-function "RefinementDefinition_1" "MaxLenInt" "Silicon"
        "SiO2" 0.001 2)
113 (sdedr:define-refinement-function "RefinementDefinition_1" "MaxLenInt" "Silicon"
        "Si3N4" 0.003 1.5)
114 (sdedr:define-refinement-function "RefinementDefinition_1" "MaxLenInt" "Si3N4"
        "PolySilicon" 0.001 1)
115 (sdedr:define-refinement-function "RefinementDefinition_1" "MaxLenInt" "TiN"
        "PolySilicon" 0.001 1.5)
116 (sdedr:define-refinement-function "RefinementDefinition_1"
        "BoronActiveConcentration" "MaxGradient" 1)
117
118
119 (sde:build-mesh "snmesh" "-a■-c■conforming" "n@node@_msh")
```

APPENDIX B
SDEVICE: CODE FOR SDEVICE SIMULATION OF TRANSFER CHARACTERISTICS OF 7NM NANOSHEET FET DEVICE

```
File {
     Grid = "@tdr@"
     Parameter="@parameter@"
     Current= "@plot@"
     Plot= "@tdrdat@"
     Output= "@log@"

  }

Electrode {
   { Name="Source"  Voltage=0.0  }
   { Name="Gate" Voltage=0.0 workfunction=@Wf@}
   { Name="Drain" Voltage=0.0 }
   }

Physics{
   * DriftDiffusion
   Fermi
   *eQuantumPotential(density)
   EffectiveIntrinsicDensity(OldSlotboom)
   *IncompleteIonization
   Mobility(
      DopingDep
      eHighFieldsaturation
      hHighFieldsaturation
      Enormal(IALMob)
      *IncompleteIonization
   )
   Recombination(
      SRH(DopingDep)
      Band2Band
      Auger
      TrapAssistedAuger
   )

   *FermiStatistics
   MultiValley(MLDA kpDOS)
   Hydrodynamic
   Piezo(
         Model(
               Mobility(
               eSubband( Fermi EffectiveMass Scattering )
               hSubband( Fermi EffectiveMass Scattering )
            )
         )
   )

}

Plot{
*--Density and Currents, etc
   eDensity hDensity
   TotalCurrent/Vector eCurrent/Vector hCurrent/Vector
```

```
54      eMobility hMobility
55      eVelocity hVelocity
56      eQuasiFermi hQuasiFermi
57
58  *--Temperature
59      eTemperature Temperature * hTemperature
60
61  *--Fields and charges
62      ElectricField/Vector Potential SpaceCharge
63
64  *--Doping Profiles
65      Doping DonorConcentration AcceptorConcentration
66
67  *--Generation/Recombination
68      SRH Band2Band * Auger
69      AvalancheGeneration eAvalancheGeneration hAvalancheGeneration
70      eSRHRecombination
71
72      Band2BandGenerationRate
73
74  *--Driving forces
75      eGradQuasiFermi/Vector hGradQuasiFermi/Vector
76      eEparallel hEparallel eENormal hENormal
77
78  *--Band structure/Composition
79      BandGap
80      BandGapNarrowing
81      Affinity
82      ConductionBand ValenceBand
83      eQuantumPotential hQuantumPotential
84
85  *--Stress related data
86      StressXX StressYY StressZZ
87      StressXY StressXZ StressYZ
88
89  *-- Quantum model related
90       eQuantumPotential
91       *SubbandEnergy
92
93  *--Thermal data(self heating)
94      LatticeTemperature
95  }
96
97  Math {
98      Extrapolate
99      Derivatives
100     Avalderivatives
101     RelErrControl
102     Digits= 5
103     ErRef(electron)= 1.e10
104     ErRef(hole)= 1.e10
105     Notdamped= 50
106     Iterations= 20
107     Directcurrent
108     Method= ILS
109 #    ILSrc=
```

```
#    set(1){
#            iterative(gmres(100), tolrel=1e-8, maxit=200);
#            preconditioning(ilut(0.0005,-1),left);
#            ordering (symmetric=nd, nonsymmetric=mpsilst);
#            options(compact=yes, verbose=0);
#     };

}

Solve {

         Coupled(Iterations=100) {Poisson}
         Coupled {Poisson Electron }
         Coupled {Poisson Electron Hole}
         Quasistationary (
         InitialStep=1e-5 Minstep=1e-12 MaxStep=0.3 Increment=2
          Goal {Name="Drain" Voltage=@Vd@})
         {Coupled {Poisson Electron Hole }}
        Save ( FilePrefix= "n@node@_Vd_@Vd@" )


         NewCurrentFile="IdVg_Vd_@Vd@_"
         Load ( FilePrefix= "n@node@_Vg_@Vd@" )
         Quasistationary (
         InitialStep=1e-5 Minstep=1e-12 MaxStep=0.03 Increment=2
         Goal {Name="Gate" Voltage= @Vg@})
         {Coupled {Poisson Electron Hole }
       }
}
```

APPENDIX C
SDEVICE: CODE FOR SDEVICE SIMULATION OF OUTPUT CHARACTERISTICS OF 7NM NANOSHEET FET DEVICE

```
File {
    Grid= "@tdr@"
    Parameter="@parameter@"
    Current= "@plot@"
    Plot= "@tdrdat@"
    Output= "@log@"

  }
Electrode {
  { Name="Source" Voltage= 0.0  }
  { Name="Drain" Voltage= 0.0  }
  { Name="Gate" Voltage= 0.0  workfunction=@Wf@}
}

Physics{
   EffectiveintrinSiDensity(OldSlotboom)
   Fermi
   *eQuantumPotential
   Mobility(
        DopingDependence
```

```
21          CarrierCarrierScattering
22          eHighFieldsaturation(GradQuasiFermi)
23          hHighFieldsaturation(GradQuasiFermi)
24          Enormal
25      )
26      Recombination(
27          SRH(DopingDep)
28          Band2Band
29          Auger
30      )
31      *FermiStatistics
32      MultiValley(MLDA  kpDOS)
33      Hydrodynamic
34      Piezo(
35           Model(
36                Mobility(
37                eSubband( Fermi EffectiveMass Scattering )
38                hSubband( Fermi EffectiveMass Scattering )
39             )
40          )
41      )
42  }
43  Physics(MaterialInterface="Silicon/Oxide") {
44   GateCurrent( DirectTunneling )
45   }
46
47  Plot {
48  *    eCurrent hCurrent AvalancheGeneration ConductionCurrent
49  *   eAlphaAvalanche hAlphaAvalanche AugerRecombination
50  *    EquilibriumPotential IntrinsicDensity
51  *    EffectiveIntrinsicDensity  Temperature ElectricField Potential Doping
52  *    SpaceCharge
53  *    eBarrierTunneling   hBarrierTunneling
54  *    conductionbandenergy valancebandenergy
55
56  *--Density and Currents, etc
57      eDensity hDensity
58      TotalCurrent/Vector eCurrent/Vector hCurrent/Vector
59      eMobility hMobility
60      eVelocity hVelocity
61      eQuasiFermi hQuasiFermi
62
63  *--Temperature
64      eTemperature Temperature * hTemperature
65
66  *--Fields and charges
67      ElectricField/Vector Potential SpaceCharge
68
69  *--Doping Profiles
70      Doping DonorConcentration AcceptorConcentration
71
72  *--Generation/Recombination
73      SRH Band2Band * Auger
74      AvalancheGeneration eAvalancheGeneration hAvalancheGeneration
75
76      Band2BandGenerationRate
```

```
77
78  *--Driving forces
79     eGradQuasiFermi/Vector hGradQuasiFermi/Vector
80     eEparallel hEparallel eENormal hENormal
81
82  *--Band structure/Composition
83     BandGap
84     BandGapNarrowing
85     Affinity
86     ConductionBand ValenceBand
87     eQuantumPotential hQuantumPotential
88
89  *--Stress related data
90     StressXX StressYY StressZZ
91     StressXY StressXZ StressYZ
92
93  *-- Quantum model related
94      eQuantumPotential
95
96  *--Thermal data(self heating)
97     LatticeTemperature
98
99  }
100
101 Math {
102    Number_of_Threads = 4
103     #Iterations = 10
104     #NotDamped  = 20
105     RHSMin = 1e-8
106     Extrapolate
107     Derivatives
108     AvalDerivatives
109     ErrRef(Electron) = 1e10
110     ErrRef(Hole)     = 1e10
111     method = pardiso
112     #ComputeGradQuasiFermiAtContacts= UseQuasiFermi
113
114     transient=BE
115     eMobilityAveraging = ElementEdge
116     hMobilityAveraging = ElementEdge
117
118     #BreakCriteria {LatticeTemperature (MaxVal =1400)}
119    -CheckUndefinedModels
120
121    RelErrControl
122    Digits= 5
123    Notdamped= 50
124    Iterations= 20
125    Directcurrent
126    #Method= ILS
127 }
128
129 Solve {
130         Coupled(Iterations=100) {Poisson}
131         Coupled {Poisson Electron }
132         Coupled {Poisson Electron Hole}
```

```
133         #Save ( FilePrefix= "n@node@_Vg_0" )
134          Quasistationary (
135          InitialStep=1e-5 Minstep=1e-12 MaxStep=0.3 Increment=2
136           Goal {Name="Gate" Voltage=@Vgs@})
137          {Coupled {Poisson Electron Hole }}
138         Save ( FilePrefix= "n@node@_Vg_@Vgs@" )
139
140 #------------- Drain Sweep at Vgs Voltage---------------#
141
142  #        NewCurrentFile="IdVd_Vg_0_"
143  #       Load ( FilePrefix= "n@node@_Vg_0" )
144  #        Quasistationary (
145  #        InitialStep=1e-5 Minstep=1e-12 MaxStep=0.03 Increment=2
146  #        Goal {Name="Drain" Voltage= @Vds@})
147  #       {Coupled {Poisson Electron Hole }}
148
149
150          NewCurrentFile="IdVd_Vg_@Vgs@_"
151          Load ( FilePrefix= "n@node@_Vg_@Vgs@" )
152          Quasistationary (
153          InitialStep=1e-5 Minstep=1e-12 MaxStep=0.03 Increment=2
154          Goal {Name="Drain" Voltage= @Vds@})
155          {Coupled {Poisson Electron Hole }
156
157          #Plot ( FilePrefix = "n@node@_NS_" Time = ( range = (0 1) Intervals= 8 )
                 NoOverwrite )
158       }
159 }
```